



How To Increase Production And Enhance Shop Productivity

This article features an innovation project at Wheel and Axle Plant (WAP) of Durgapur Steel Plant on designing, fabrication and installation of a low-cost, electronic wheel counter with a large LED display at wheel forge and wheel mill area to motivate people for higher production and enhance shop productivity



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It is said that India moves on DSP (Durgapur Steel Plant) wheels as Durgapur Steel Plant of the state-owned steel giant Steel Authority of India (SAIL) is the primary supplier of forged wheels and axles for Indian Railways. Wheel forging area is one where a wheel takes birth from its mother steel block and undergoes the infantry stage of getting the desired primary shape for its further journey. Hence, it is an important area in the wheel manufacturing process, where all sorts of precautions and care are required for

the smooth birth and growth of the wheel before it is sent for machining for finishing operations.

Moreover, total wheel production target has to be achieved while maintaining critical aspects. The entire production area has inter-related sub-sections or workstations where line production of the forged wheel takes place.

Sub-sections or workstations under the production line of forged wheels are furnace, forging press, wheel mill, dishing press, stamping press and so on, as shown in Fig. 1.



Wheel blocks cut from special steel ingots



Forging of hot block



Operation in wheel mill



Heat-treated forged wheels



Machined wheels



Finished wheels for despatch

Fig. 1: Process of making wheels